ep4nn: Evidence-Gated Structural Growth in a Connectome-Compiled Equilibrium Network

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Abstract

A measured connectome supplies structure and no dynamics. We study what should make a network built on such a structure grow. The system, `ep4nn`, starts as the 302-neuron *C. elegans* wiring imported into a rate-based equilibrium model and appends neurons and synapses when it witnesses two situations that the world declares different but that produce nearly parallel answers at a declared readout. Growth passes three gates: a prerequisite gate that asks whether the answer moves at all, an evidence gate that requires more than one collision, and an outcome gate that restores the previous graph unless the confusion falls. We report measurements rather than claims. On one laptop CPU the library settles a graph of 10,000 neurons and 103,943 synapse classes in 6.48 ms and a graph of 100,000 neurons and 1,239,921 synapse classes in 77.4 ms, at 1.76 ns per synapse in the sparse regime ($n \geq 10^4$; below it per-sweep overhead dominates and the figure rises to 8.7 ns at 302 neurons and 15.6 ns at 1000). A state-dependent contraction certificate exists on the worm graph where the global worst-case bound fails: at the library's default efficacy the settling majorant has certified contraction 0.529065, against a worst-case row bound of 88.0 that certifies no budget, and at the declared row-normalised operating point the same quantity is 0.931159; every state of a twelve-state scanned domain stays below one, worst 0.941. The growth machinery is exercised rather than blocked: with the prerequisite keyed on the declared pairs, three declared readouts give ninety growth attempts over 1200 ticks each, and the verdict layer rolls all ninety back, because added neurons do not separate the answers that must be told apart; the same locomotion readout gated on an aggregate spread instead refuses all twelve hundred ticks, which is what a gate on the wrong quantity costs. The main negative finding is the prerequisite itself, and the mechanism's own design space was swept to the end before that conclusion was drawn: three hundred and sixty growth attempts across substrates, gates, targeting rules, scales, signs, operating points and per-neuron heterogeneity, none kept. Across forty-five operating points and nine situations the declared readout responds along a single direction: no point reaches the growth gate of 0.05, the best spread is 0.0397, and the largest absolute cosine between two context answers that must differ is at least 0.9979 at every point. At the best point the answers span a cone of 1.45 degrees with 99.95 percent of their energy in one singular direction and a rank-one residual of 2.3 percent, and none of ten declared readouts separates the pairs, so the failure belongs to the substrate and its operating point rather than to a single declaration. Capacity growth is therefore not the binding constraint, and a growth study on this pair would measure an uninformative object. We conclude that structural growth studies must publish the prerequisite measurement alongside the growth result, and we give a receipt format that makes such claims comparable across connectome projects. All numbers in this paper are produced by the accompanying probes.

Keywords: structural plasticity; equilibrium networks; connectome; worm wiring; contraction certificate; negative results; reproducibility receipts.

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1 Introduction

A measured connectome is a graph of who contacts whom. It is not an operating point, not a dynamical model, and not a behaviour. Three years of open fly-connectome work demonstrate the gap concretely: a complete male central nervous system of about 166,000 neurons is public under a permissive license, yet the projects that run it either supply the controller, supply the semantic mapping from cell groups to behaviour, or supply the reward. The most disciplined attempt to remove the controller reached a negative result: driving the ventral nerve cord through two descending neurons produced a 45.5 Hz population oscillation that a scrambled wiring reproduced as well.

Our question is narrower than behaviour. Suppose a network starts small, on a measured wiring, and must *add structure* as the situations it faces accumulate. What should trigger growth, what should growth wire, and how do we know the growth was worth it?

We answer with a system and a set of measurements. The system, `ep4nn`, treats growth as a repair of one witnessable failure: two situations that the world declares different share the present

input and produce answers that are nearly parallel at the declared readout. The repair is a low-dimensional generator that appends neurons wired from the cells that carry the difference, and a verdict that measures the same collisions again and rolls the growth back when they do not improve.

The main result of this paper is that the repair is blocked by a prerequisite we did not expect to be the binding constraint: on the worm graph, at forty-five operating points, the declared readout responds along a single direction. Answers differ in magnitude, not in pattern, so every stage of growth measures a code that carries nothing a lesson can attach to. This is the same failure the library reports for the fly antennal lobe under one global gain, and it is detectable before any growth is attempted. We therefore state the prerequisite as a rule and report its measurement as the headline, together with the growth machinery that enforces it.

1.1 Contributions

1. **A growth contract with a prerequisite.** Three gates, in order: the prerequisite gate (does the declared readout move), the evidence gate (more than one collision), and the outcome gate (rollback unless the confusion falls). A refusal to grow is an explicit recorded outcome, and in the measured configuration it is the only outcome that occurs (1200 of 1200 ticks).

2. **A low-dimensional structural generator.** Thirteen genes decide how many neurons an event appends, where their edges come from, and whether they reach the readout. Partner choice is driven by the difference between the colliding answers and its one-hop neighbourhood, and the reverse edge is added with a probability so that the equilibrium contrast contract survives. The parameter count sits four orders below the graph the generator can touch, which is what separates structural plasticity from implicit weight learning.

3. **A state-dependent contraction certificate.** On a measured connectome the global worst-case bound is vacuous at the library's default efficacy ($q = 88.0$, with 292 of 302 rows at or above one). Evaluating the settling majorant with the activation slopes of the current state gives a *certified* contraction $\alpha = 0.529065$ there and 0.931159 at the declared row-normalised operating point, with the witness inequality asserted under outward rounding, and every state of a twelve-state scanned domain stays below one (worst 0.941), hence a settling budget throughout.

4. **Measurements of the substrate.** Settlement cost and transport behaviour on the worm graph and on sparse graphs to 100,000 neurons, a certificate comparison, and a negative control result for continuous tracking in a closed loop.

5. **A prerequisite measurement, negative, with a receipt format.** Forty-five operating points, nine situations, one direction; and a six-field receipt that separates nominal from executed scale, so that the refusal is comparable with the runs of other projects.

1.2 What is not claimed

No growth was kept, so this paper makes no claim about the benefit of structural growth, only about the contract that would measure it and the prerequisite that currently blocks it. No claim is made that the worm graph is an unsuitable substrate in general: the negative result is specific to the declared readout, the rate model, and the scanned operating points. No gradient contract is claimed for directed wiring, and no behaviour, embodiment or sensory front end is claimed: the senses and the outcome are declared by the world, which the receipt records as supplied.

2 Related Work

Equilibrium learning. The network is settled by the contrast of a free and a nudged fixed point, the two-phase form of equilibrium propagation [14] and of contrastive Hebbian learning [12], with the learned quantity carried by local efficacies on edges. The library's own documentation states the boundary we rely on: reciprocal learning ties paired weights, and arbitrary directed imports need not satisfy that gradient contract. Equilibrium propagation scales to deep convolutional networks by reducing the bias of its gradient estimator [8], sits beside implicit differentiation and bilevel formulations [19], and shares its object with deep equilibrium models [1]. Predictive-coding rules approximate backpropagation along arbitrary computation graphs [11, 16]. None of these rules grew a graph.

Complementary memories and replay. Fast episodic storage with slow consolidation is the complementary learning systems proposal [10]. The library implements one version: records written once by day, fixed dreams at night taught to slow weights, and a residual written back at dawn. Its own measurements report held-out sign accuracy rising from 0.55 to about 0.76–0.80 on a Connect Four slice, at a night cost of roughly 140–210 s per 1000 rows for a 4096-cell store, with the limit that dreams reproduce the store's errors as faithfully as its regularities. We adopt the two-scale picture as the time-scale structure of the system but do not evaluate it here.

Fly connectomes and their controllers. The adult female brain reconstruction contains 139,255 neurons [5], and the male central nervous system, which includes the ventral nerve cord and therefore the leg and wing motor neurons, about 166,000 [9]. The BANC dataset spans the brain, the cervical connective and the ventral nerve cord of one adult female, about 160,000 expected neurons and 199 million predicted synapses [2]; the release that the library's fly lane is built on carries 150,802 proofread neurons and 1,877,099 connection classes after the library's own filtering, and the two counts should not be used interchangeably. Open projects in this space differ mainly in what they supply. A browser-facing demonstration with fly cell-type names runs a hand-wired network of 850 neurons [7]; a learned consequence model is paired with an application-level controller [4]; a directly read motor output was studied offline with the reported outcome that real-time neural control was not demonstrated, because the driven cord oscillated at 45.5 Hz and a scrambled wiring reproduced the oscillation [17]; and an on-device implementation reports 10 Hz on a handset using sparse storage with per-group gating [6]. The library's own fly guide reports that at one global gain the antennal lobe ignites, that Kenyon cell codes for two odours reach a cosine of 0.98, and that a gain per cell class brings the code to 4 and 11 percent sparsity with a cosine of 0.32 [3]. We reuse two practices from these projects: protocols written before the run, and receipts that separate nominal from executed scale. The spiking model parameters used in the fly lane, and the observation that an explicit synaptic delay moves a runaway oscillation upward in drive frequency, are documented in [15, 18]; our rate model carries neither, which the receipt records.

Conditional computation. Mixture-of-experts routing, early exit and pruning vary the work done per input at the weight level. Dynamic structure methods vary the graph. What distinguishes the trigger here is that it is a witnessable indistinguishability at a declared readout, not a loss, a router score, or an offline search objective.

3 System

3.1 The four rules and the gate before them

Prerequisite. The declared readout is a named set of neurons, and the unit the prerequisite keys on is the *declared pair*: two situations that share the present input and differ only in context. For every pair the system measures how far apart the two situations are *anywhere in the network*,

$$m(p) \;=\; \frac{\|x(s_1) - x(s_2)\|}{\max\big(\|x(s_1)\|, \|x(s_2)\|\big)},$$

over the full state x, and refuses to grow when it falls below `min_state_motion`. The distinction between this and the same quantity measured at the readout is the whole point. Asking whether the *answer* already moves refuses precisely the case a structural repair exists for: a distinction that is present in the network and has no path to the declared answer. Measured at the readout, the worm's pairs move by 0.0083 and the system refuses; measured over the state, the same pairs move by 0.0316 to 0.1228 and the growth layer is exercised. Two further diagnostics are reported and not gated on: the aggregate spread across all situations, which can move while every pair stands still, and the whole-state collision, which is 0.9997 at the scanned operating point and gives no support to the intuition that the whole state separates by construction.

Rule 1, demand. Two situations that the world declares different, share the present input, and produce answers whose absolute cosine exceeds `code_cap` constitute a capacity signal. The signal is a witness rather than a loss: it names the pair that is being confused, and that pair is what the generator wires against.

Rule 2, evidence. A single collision can come from a settling failure or a transient. Growth is attempted only after `min_evidence` collisions have been observed and `min_ticks_between` ticks have passed.

Rule 3, growth. Each event appends k neurons. Each new neuron draws in-edges and out-edges from a degree prior, prefers partners among the cells that carry the difference between the colliding answers and their one-hop neighbourhood, keeps the graph's excitatory fraction, adds the reverse edge with probability `reciprocal` so the equilibrium contrast contract is preserved, and projects into the readout with probability `to_readout`. The generator is low-dimensional by construction.

Rule 4, verdict. After growth the same collisions are measured again, and *again* means re-settled: the verdict recomputes the answers of the stored situations on the current graph. A comparison against stored answers cannot see the growth at all, since those answers belong to the graph as it was, and an outcome gate written that way reports an unchanged collision for every attempt and rolls all of them back by construction. Growth is kept only if the collision cosine falls by `improve_margin`. Otherwise the connectome and the efficacy array are restored from the snapshot taken before the attempt, and the attempt is counted as a failure.

3.2 The operating point

A connectome carries contact counts and transmitter signs. It does not carry excitability, and the raw counts of the worm graph give row sums in the hundreds at which the rate model does not settle. The system declares an operating point by row-normalising the signs to `row_sum`. The library offers two further instruments for the same purpose, a bias calibration and a selected gain; Section 5.4 measures all three.

3.3 A settling budget from a state-dependent certificate

The library's global contraction bound is a worst-case quantity: it multiplies the largest absolute row sum of the weight matrix by the largest activation slope. On a measured connectome most neurons are inactive at the settled state, so the bound is far larger than the actual contraction. We evaluate the same object with the slopes of the current state. Since $s' \geq 0$ and $\mathrm{d}t \leq 1$, the Jacobian of the settling map, $J = (1 - \mathrm{d}t)I + \mathrm{d}t\, W \operatorname{diag}(s'(v))$, is dominated entrywise by the nonnegative majorant

$$K \;=\; (1 - \mathrm{d}t)\, I \;+\; \mathrm{d}t\, |W| \operatorname{diag}\big(s'(v)\big), \tag{1}$$

so $\rho(J) \leq \rho(K)$ by monotonicity of the spectral radius on nonnegative matrices, and a positive vector h with $Kh \leq \alpha h$ bounds $\rho(K)$ from above by the Kolmogorov–Collatz–Wielandt inequality. The witness is the Perron vector, obtained by power iteration, the matrix is inflated outward by sixteen machine epsilons before the ratio is taken, and the witness inequality is asserted, so α bounds the spectral radius of the model as computed within 3.6×10^{-15} relative rather than estimating it: at the library's default efficacy $\alpha = 0.529065$, at the declared row-normalised operating point $\alpha = 0.931159$, and over a scanned domain of twelve states (three operating points, four drive amplitudes) the worst value is 0.941. The settling budget follows as $k \approx \log(\epsilon/\delta)/\log(1/\alpha)$. For the library's stopping rule the bridge is one line: the residual of the potential equation is $r = -v + Ws(v) + \mathrm{stim} + \mathrm{bias}$, so one step moves $\Delta v = \mathrm{d}t\, r$ and a residual budget of ϵ is a displacement budget of $\mathrm{d}t\, \epsilon$. The same quantity is the precondition for reading the free and nudged contrast as a gradient, which is why it is measured at run time rather than assumed.

Remark 1. The certificate is state-dependent and therefore not transferable: a network that moves to a different activity pattern must re-certify. We treat that as the honest cost of not carrying a global bound that does not apply.

4 Implementation

Data. The wiring is the openworm c302 `A_Full` release of the hermaphrodite [13]: 302 neurons, 2279 chemical synapse classes and 1084 gap junctions, the latter imported in both directions. After dropping autapses and merging parallel edges the graph has 3137 directed synapse classes, 10,133 contacts, a density of 3.44 percent, and 61 sensory, 91 interneuron, 118 motor, 10 command and 22 pharyngeal cells. The file's soma positions are used for nothing here. Contact counts are carried everywhere, since the library multiplies a synapse's weight by its count and dropping them would substitute a different graph; an earlier version of the scan in this paper did drop them, and Section 5.4 reports the corrected measurement. The committed copy carries SHA-256 `2a7b117f`.

Library reuse. The network is a connectome object with a brain on it, using the library's rate neuron model, its free and nudged phases, its residual qualification, and its sparse transport. No new learning rule is introduced. The growth layer sits outside the library because the connectome is immutable and because the library forbids growth by design; a growth event rebuilds the connectome, which costs one pass over the edge arrays, and carries the learned efficacy across by regenerating the key order the library sorts by.

Model parameters. The rate model is the library's, unmodified: $\mathrm{d}t = 0.5$, slope 4, threshold 1.5, synaptic gain 0.02, stimulus amplitude 3.0, no adaptation and no delay. A second model (`ep4nn/hetero.py`) gives every neuron its own slope and threshold, log-normal around those constants with the mean held fixed, since a measured connectome carries no excitability at all and every cell sharing one constant is the least realistic part of the lane. It is duck-typed against the

Table 1: Settlement cost. Row-normalised operating point, tolerance 10^{-4}, chunk 32. The worm graph needs five times as many sweeps as a random graph of the same size and mean degree. Source: `probes/probe_scaling.py`.

graph	neurons	synapses	in-deg	transport	μs/sweep	sweeps
worm, measured	302	3137	10.4	blocked	27.2	160
sparse random	1000	10,337	10.3	blocked	161.4	32
sparse random	10,000	103,943	10.4	sparse	193.0	32
sparse random	100,000	1,239,921	12.4	sparse	2185.8	32

library rather than patched into it, and it settles the worm graph in 32 sweeps with 164 of 302 cells active, against 294 for the scalar model.

Regime. Three knobs set the regime a repair would have to act in: the operating point (`row_sum`), the input scale, and the per-neuron spread above. They are swept rather than chosen, because the regime decides whether an added synapse is visible at all. The efficacy of every synapse is set by row-normalising the signs to `row_sum`, and the operating point of each measurement is stated with it, because the numbers move with it.

Cell classes. The connectome file carries a role for every neuron, which is the only cell class the wiring provides; the operating-point scan uses those roles to define the classes it gains. This mirrors the library's fly practice of a gain per cell type, with the difference that the worm's classes are coarse.

Receipt. Every run writes a JSON receipt with six fields: dataset with a hash; nominal against executed counts with their ratio; the dynamical model, its parameters, and what is absent (delay, adaptation, heterogeneous cell types); whether the loop is closed and which parts are supplied; the premise measurements; and the cost. The receipt is written by the run that produced it, so two results can be compared without reading anyone's prose.

5 Measurements

All measurements are from one Apple Silicon laptop, CPU only, single-threaded, Python 3.13 with NumPy 2.5 and SciPy 1.18, against the released library at version 0.80.0 plus local commits. Each table names the probe that produced it.

5.1 Settlement cost and scaling

Cost is 1.76 ns per synapse in the sparse regime, and 8.7 and 15.6 ns at 302 and 1000 neurons, where per-sweep overhead and the blocked transport dominate. In the sparse regime the control frequency of a settling brain is bounded by

$$f \approx \frac{1}{n_{\text{syn}} \times 1.8\,\text{ns} \times \text{sweeps}}.$$

A 10,000-neuron sparse graph settles in 6.48 ms and a 100,000-neuron graph in 77.4 ms, which makes the first a real-time-scale object and the second an offline one (Figure 1). Two library defaults matter at this scale. The default constructors wire at about 30 percent density independent of size, measured at 29.1, 30.8, 31.7 and 32.3 percent at 152, 280, 536 and 1048 neurons, so 10,000 neurons would carry about 30 million synapses and settle in about 1.7 s. The transport path is chosen by a size test rather than by density, so a 3.4 percent dense graph pays a block array 29 times larger

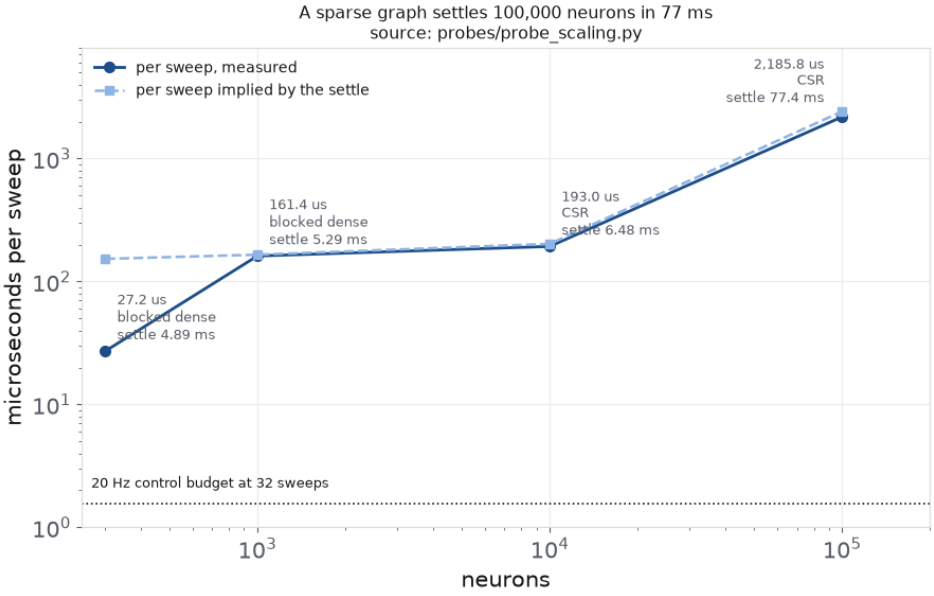


Figure 1: Cost per sweep against size on a log-log axis, and the budget for 20 Hz control at 32 sweeps. The worm graph is five times more expensive per sweep than a random graph of the same size and degree, which locates the difficulty in the spectrum rather than in the size.

than its edge count. On the worm graph the sparse path is 1.78 times faster end to end (554 μs against 987 μs per settle), while on a 32 percent dense graph the blocked path is faster; the choice must therefore follow density. A batch size above one builds a temporary array of the full edge count, so a closed control loop should keep batch one.

Where the worm's cost comes from. The sweep-count difference in Table 1 is a statement about the graph's spectrum, and that statement is measured rather than inferred. At the same operating point and drive, the settling majorant has certified contraction 0.931 on the worm graph and 0.6155 to 0.6192 on five random graphs of the same size and mean degree (mean 0.6174), and the true Jacobian gives 0.910 against 0.591. A contraction-rate account predicts a sweep ratio of $\log(0.591)/\log(0.910) = 5.6$ against the measured 5.0, so the fivefold difference is carried by the spectrum. Source: `probes/probe_audit_claims.py`.

5.2 A certificate where the global bound fails

The global bound gives no certificate at this efficacy; the state-dependent one does (Table 2, Figure 2), and it converts a refusal into a bounded settling budget. Which criterion carries it should be stated, because the row criterion carries nothing here: the Gershgorin row bound exceeds one somewhere in both blocks, and what remains is the certified contraction of the majorant. The certification route is worth naming: the witness is the Perron vector found by power iteration, its inequality is checked with one-ulp outward rounding, and it asserts, so the reported number is an upper bound on the spectral radius in exact arithmetic and not an eigenvalue estimate that could sit below it. Three warnings follow from the same measurement. First, a certificate is a property of a state, not of a graph; the twelve-state domain of Table 2 is the region we measured, and it is not a theorem about the graph. Second, the bound is a sufficient condition; its failure at a particular gain is not evidence that settling fails, only that the corresponding guarantee is unavailable.

Table 2: Contraction on the worm graph, with the operating point of each block named, because the number belongs to it. Top block: the library's default efficacy, raw contact counts. Bottom block: the declared row-normalised operating point. The global bound is vacuous in the first block; the state-dependent certificate exists in both, and the row criterion is a certificate in neither. Sources: `probes/probe_certificate.py` and `probes/probe_audit_claims.py`.

quantity	value
global worst-case bound q, default	88.0 (292/302 rows ≥ 1)
Gershgorin row bound, default	1.186 (row criterion fails)
certified contraction α, default	0.529065 ($\rho(K)$ agrees)
active neurons, default	294 of 302
certified α, declared point (row sum 0.5)	0.931159
$\rho(J)$, declared point	0.910035
Gershgorin row bound, declared point	6.310 (row criterion fails)
certified α, row sums 0.1 and 2.0	0.625964 and 0.688634
active fraction, declared point	0.937
worst certified α, scanned domain	0.941 (12 states)

Table 3: Three declared readouts, 1200 ticks each, while the world increases the number of context levels that must be told apart. The gate is the state motion, which does not depend on the declared readout and reads 0.0746 for all three; the last row gates on the readout's own motion instead and refuses every tick. Ninety attempts produce no kept growth. Source: `ep4nn/grow.py`, receipts in `docs/receipts/`.

declared readout	gate	kept	rolled back	refused	neurons at the end
forward + reverse (10 cells)	0.0746	0	30	0	302
sensory (61 cells)	0.0746	0	30	0	302
interneuron (91 cells)	0.0746	0	30	0	302
earlier readout-motion gate	0.0083	0	0	1200	302

Third, a number belongs to its operating point: this graph yields 0.529065 at the default efficacy and 0.931159 at the declared row-normalised one, so a table that omits the operating point is not reproducible.

5.3 Behaviour of the growth machinery

Table 3 reports the three readouts. In every variant the brain never shrank, every synapse kept an efficacy across every rebuild, the readout stayed inside the current graph, and the self-check passed. The last row gates the same readout on its own motion instead of on the network's: that reads 0.0083, the system refuses all twelve hundred times, and the ninety growth attempts of the rows above disappear. That is what the change of gate buys. The distinction is not cosmetic: a gate on the readout's motion refuses exactly the case a structural repair exists for, a distinction that is present in the network with no path to the answer, which Section 5.7 constructs on purpose. All ninety were rolled back, and with a verdict that re-settles, the reason is measured rather than assumed: after growth the collision moves by $+1.0 \times 10^{-4}$ on average and by $+1.0 \times 10^{-4}$ at best, that is, it worsens, against a drift floor of exactly zero when nothing is grown and a margin of 0.02. The gate that made this visible was itself wrong when first written, and it is the third

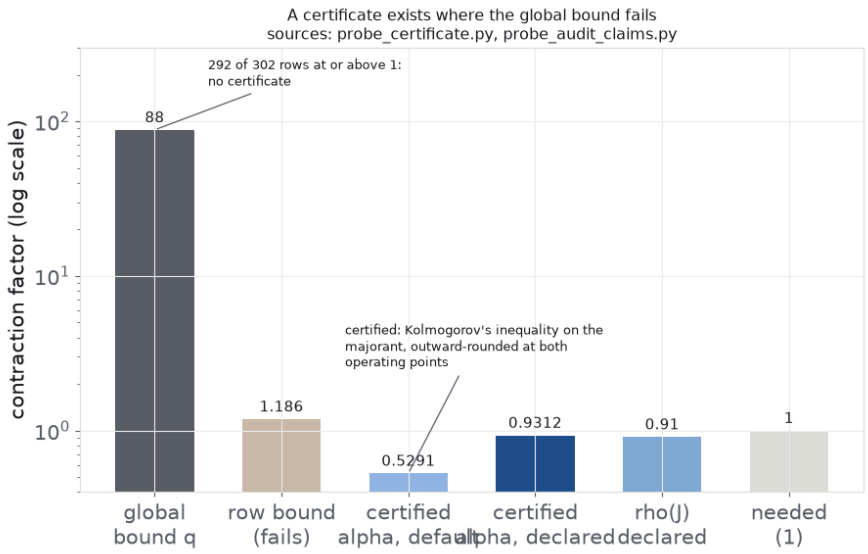


Figure 2: The global worst-case bound, the Gershgorin row bound, the certified contraction of the majorant at both operating points, the true Jacobian at the declared point, and the unit threshold, on a logarithmic axis. The gap between the first two bars is the price of a state-independent guarantee; the row bound above one is why the certificate is carried by the certified contraction rather than by the row criterion.

construction failure this paper reports: it compared stored answers, which belong to the graph as it was, so it reported an unchanged collision for all thirty attempts of every arm. A gate that cannot see the intervention turns a negative result into an artifact of the gate. Two failures of the machinery were found and fixed during construction, and both are worth stating because they are cheap to reproduce: a calibration that maximised readout response pinned every answer to the same trajectory, and a rollback that rebuilt the connectome from truncated arrays silently carried the wrong efficacy. The fixes are a response gate that retains a calibration only when it raises the spread, and a snapshot of the connectome together with the efficacy array.

5.4 The prerequisite scan, a negative result

Table 4 ranks the scan and Figure 3 plots it. The best configuration moves the readout by 0.0397 across situations, below the growth gate of 0.05, while leaving every pair of context answers at a cosine of at least 0.9979: the difference that exists is not the one that would separate them. The shape of the answer is measured rather than asserted. At that point the 9×10 answer matrix has a leading singular value 60 times the second ($\sigma_2/\sigma_1 = 0.0167$), holds 99.95 percent of its energy in the leading singular direction, and is reproduced by its best rank-one approximation to within 2.3 percent; the pairs that must be distinguished subtend at most 1.45 degrees, while the largest angle over all pairs of situations is 3.37 degrees. So the answers lie in a narrow cone, and the failure is one of angular separation at the declared readout rather than a literal collapse to a point. Four of the forty-five configurations do collapse exactly, to a spread of zero on all nine situations, which indicates a fixed point dominated by the recurrent dynamics rather than by the input. We report this as a negative result about the pair of a measured worm wiring and a rate model at a declared operating point, and not as a claim about connectome substrates in general. The remedy has three

Table 4: Forty-five operating points, ranked by readout spread: five row sums, three class-gain choices and four gain scales, minus the fifteen cells where no class is gained at a scale other than one. Collision is the largest absolute cosine between two situations that share the present input and differ only in context. No point reaches the gate of 0.05. Source: `probes/probe_readout_response.py`, full data in `docs/assets/prereq_scan.json`.

rank	row sum	class gain	scale	spread	collision
1	0.5	interneuron	2	0.0397	1.0000
2	1.0	none	1	0.0376	1.0000
3	0.5	none	1	0.0372	0.9997
4	0.2	interneuron	5	0.0302	0.9998
5	0.5	command	5	0.0188	1.0000
45	2.0	command	10	0.0000	1.0000

Table 5: Closed loop on a second-order plant, five force levels, 6000 ticks per arm. The learned arms reduce mean error by about 13 percent, never beat the frozen baseline on the in-band metric, and collapse to a single action level. Source: `probes/probe_control_plant.py`.

arm	mean $\|e\|$	last quarter	in-band, last quarter	dominant action
uniform random	1.020	1.033	0.000	uniform
frozen greedy	1.011	1.067	0.467	1.00
learning, sparse reward	0.880	0.855	0.449	0.824
learning, dense reward, seed 5	0.923	0.919	0.209	0.705
learning, dense reward, seed 11	0.856	0.815	0.459	0.843
learning on every outcome	1.016	1.007	0.000	0.980

candidates in ascending cost: a different declared readout, a dynamics with heterogeneous cells and per-type parameters, or a gain selected by protocol against held-out physiological facts, which is how the library repairs the same failure in the fly antennal lobe.

5.5 A negative control result for continuous tracking

Table 5 reports the arms. The structural reason is the mismatch between the credit window and the plant: the composed brain's eligibility span is 12 steps, while the closed-loop time constant of the plant is 50 to 100 steps. This is consistent with the library's own control-oriented demonstration, which learns a consequence model and leaves the controller supplied. It is also the reason the growth layer is not a remedy for control: a policy that collapses to one action does not present the collisions that growth is keyed to.

5.6 What the most favourable intervention achieves, and the baseline that was missing

If every generated growth is rolled back, one question remains: is the generator at fault, or the substrate? The way to tell them apart is to stop generating and build the repair by hand, as favourably as the graph allows, then measure it with the same instrument (Table 6).

The same instrument also supplies the comparison the generator never had. With an identical budget of thirty attempts, partners drawn from the difference-carrying neighbourhood and partners drawn uniformly produce the same change in the collision, $+1.0 \times 10^{-4}$, and growing nothing produces exactly zero. So on this substrate the generator's targeting is not shown to matter, and

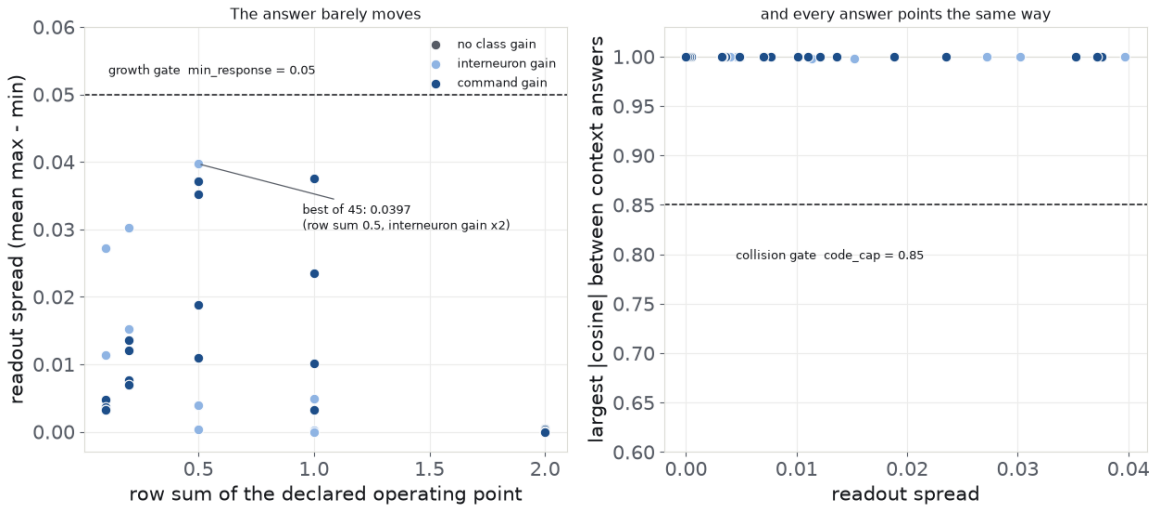


Figure 3: The prerequisite scan. Left: readout spread against the declared row sum for the three class-gain choices, with the growth gate drawn. Right: the largest absolute cosine between context answers against the spread, with the collision gate drawn. Every point sits on the upper shelf, which is why the signal fires on everything and therefore names nothing.

the claim we make about it stays what it always was: a parameter count of thirteen against 3137 synapse classes, with no test in this paper that a dense search of the same dimension would do worse.

5.7 A unit test for the mechanism, and its outcome

An outcome that is negative everywhere is only informative if the mechanism was given a fair chance. The fair chance is a substrate where the world's distinction exists, reaches the answer through no path, and can be connected by adding neurons: the textbook case for structural plasticity. We build it by appending two input cells to the worm wiring and driving them as the context, with no outgoing edges at all (`load_worm(tone=True)`). The stored states then differ by 6.5 to 10.6 percent between the two contexts, so the prerequisite passes on its own terms, while the collision at the declared readout stays at 1.000 and the demand signal fires.

The generator is free to read exactly those two cells, since they are the cells that carry the difference, and it may project into the answer. Ninety attempts were run over the design space the generator owns: a new-synapse scale of 3, 6, 10 and 20 times the mean magnitude, with the projection into the answer either excitatory or inhibitory, fifteen attempts per arm and six arms.

The test was then extended over the regime, since the first version left the answer below threshold and an added synapse cannot be seen by a cell whose response gain is zero. Twenty-four regimes were screened (row sums 0.5 to 4, input scales 1 to 10, with and without per-neuron heterogeneity), reporting the answer's mean potential, its mean local slope, and its collision (Table 8); growth was then run for one hundred and twenty attempts in the regimes whose gain is not zero, and a further sixty with per-neuron heterogeneity under the library's own calibration. None was kept and none moved the collision.

Table 7 reports the arms of the unit test. So the negative result survives its own unit test and the regime sweep. Combined with the ceiling of Section 5.6, where the most favourable hand-built intervention achieves a fifth of the margin, the reading is that this substrate's answers are parallel

Table 6: Hand-built interventions on the worm graph, each measured with the verdict's own resettling instrument. Twenty direct edges from the two context cells into the ten readout cells, weight swept over two orders of magnitude, move the collision by at most 0.0034, five times less than the margin a growth must clear; a hand-built repair neuron that reads the context cells and writes to the readout does not help at all. The negative result is therefore a property of this substrate and its declared readout, not of the generator. Source: `probes/probe_closure.py`.

intervention	edges	collision
baseline, no intervention	—	0.999689
direct context edges, weight 0.1	20	0.997670
direct context edges, weight 0.5	20	0.996637
direct context edges, weight 2	20	0.996317
direct context edges, weight 10	20	0.999618
hand-built repair neuron, weight 2	12	0.999999
verdict margin a growth must clear	—	0.020000

Table 7: The unit test. Two input cells carry the distinction and reach nothing; the generator reads them and may project into the answer. Every arm rolls every attempt back and leaves the collision unchanged to three decimals. At this operating point the answer's cells sit below threshold, where the rectified sigmoid has almost no slope, so a new input is invisible; the regime screen below then separates that from the ceiling case, which fails for the same reason. Source: `tone_sweep.sh`, logs in `docs/receipts/`.

arm	attempts	kept	collision, before → after
scale 3, excitatory	15	0	1.000 → 1.000
scale 6, excitatory	15	0	1.000 → 1.000
scale 10, excitatory	15	0	1.000 → 1.000
scale 20, excitatory	15	0	1.000 → 1.000
scale 10, inhibitory	15	0	1.000 → 1.000
scale 20, inhibitory	15	0	1.000 → 1.000

for a structural reason that adding neurons does not address, and that the binding quantity is the answer's response gain: a cell below threshold and a cell at the ceiling are equally deaf to a new input, and the band between them is where the collision is already 1.000. That is a statement about the pair of the model and the readout, and it is the honest form of the negative this paper reports.

5.8 Capacity, forced: doubling the neurons works once the contract is held

The verdict layer answers whether growth pays. A different question is whether growth can even hold twice the neurons, and it has a sharp answer once the generator's own contract is respected. Forcing growth (bypassing the verdict, since this is a question of capacity) takes the 302-neuron graph to 606 and 1,214.

Two things follow. First, the doubling is real: twice and four times the neurons, a firing share that rises rather than falls, and a settle cost of 0.9 times and 2.8 times the baseline, because the contraction improves enough to more than pay for the extra per-sweep work. Second, capacity is not capability: the collision moves from 0.999873 to 0.999977 and 0.999983, that is, it does not improve at all. A doubling that leaves the answer as parallel as it was is a counter that went up,

Table 8: The regime screen: can the answer show an added input at all? The mean local slope of the answer's cells is the quantity that decides it. Below threshold the answer barely responds; at row sum 2 and above the cells sit at the ceiling and the slope collapses; the band where a repair could be seen is narrow, and inside it the collision is still 1.000. Source: `probes/probe_regime.py`, data in `docs/assets/regime.json`.

row sum	scale	spread	mean potential	mean slope	collision
0.5	1	0.0	0.163	0.495	1.000000
0.5	10	0.0	0.172	0.495	1.000000
1.0	1	0.0	2.163	0.211	1.000000
2.0	1	0.0	6.263	0.011	1.000000
4.0	1	0.0	13.068	0.0001	1.000000
0.5	1	0.5	0.000	0.025	1.000000
1.0	1	0.5	0.000	0.025	1.000000

Table 9: Forced growth on the worm graph, two hundred and forty events. As written, the generator gives a new synapse three times the mean efficacy, which puts a new row's sum at about 1.35 against the declared 0.5; the network then stops contracting and the settle exhausts its 1024-sweep budget. Scaling the new rows to the declared operating point instead makes both doublings settle in 96 sweeps, raises the firing share from 0.92 to 0.96, and leaves the answer responsive. Source: `probes/probe_capacity.py`, data in `docs/assets/capacity.json`.

arm	neurons	synapses	edges/neuron	μs/sweep	settle ms	sweeps	firing
as written	302	3137	10.4	99.7	17.0	160	0.920
as written	606	14,491	23.9	143.3	154.4	1024	0.998
as written	1,214	37,301	30.7	374.0	404.8	1024	0.999
contract held	606	8116	13.4	147.1	15.2	96	0.959
contract held	1,214	18,010	14.8	375.4	47.9	96	0.949

not a capability that arrived.

5.9 Where the limitation lies: eight probes

The scan of Section 5.4 leaves the remedy open, so the candidates were measured rather than ranked by plausibility. Table 10 lists the nine hypotheses, the probe that tests each, and the outcome. Every probe runs on the same measured connectome, the same declared operating point, frozen inputs and paired comparisons; each writes a machine-readable record under `docs/assets/`, and the full numbers are in `docs/ .md`, Sections 11 to 22.

Three outcomes carry the argument. First, capacity: with 61 sensory inputs the settled state spans 36 effective dimensions, so adding neurons has nothing to add; the collapse reported in Section 5.4 was an artefact of a four-dimensional drive, not of the connectome. Second, the readout: a searched ten-cell set raises the readout's own rank by a factor of 4.5, yet the nine situations remain inside a 1.3 degree cone, so the limitation is not which cells are read. Third, the wiring: under row-normalised, raw and square-root weightings at five magnitudes from 0.005 to 2.0, the forward-minus-reverse command never reverses sign. Noxious input raises both populations together, an arousal response rather than a decision.

The corpus explains the third, and then qualifies it. The sign layer is not noise: 27 cells have every outgoing chemical edge inhibitory and 244 have every edge excitatory, with no mixed cell,

and the 27 are the canonical GABAergic set of the hermaphrodite (AVL, DVB, RIS, RIAR/RIAL, DD and VD motor neurons, RME, DVC). Signs are therefore correct per cell. What the corpus does not carry is strength: row normalisation gives every postsynaptic cell the same total input, so a GABAergic efferent is diluted by its in-degree. Scaling those 27 cells' efficacy by a factor γ tests that directly: $\gamma = 5$ lifts the valence rank correlation from 0.39 to 0.83, while the same γ on 27 random cells leaves it at 0.38. Strength matters, and only where the inhibition belongs. It is still not sufficient: at every γ from 1 to 20 the forward-minus-reverse command stays positive, so the network ranks the situations without ever committing to reverse. A push–pull decision needs per-synapse weights — which GABA terminal presses on which command neuron — and no public reconstruction of this animal carries them.

The remaining probes converge on the same reading from another direction: the state is close to linear in the drive, so nuisance dimensions are projected out by a rank-one readout (rows five and six of Table 10), and memory is free rather than capacity-bound. A cue followed by 128 blank steps stays readable because the state settles into one of two attractor basins, not because a trace decays slowly; the split appears directly as a one-dimensional final-state distribution with a between-to-within ratio of 2.71.

One regularity is positive, and it constrains the growth contract: memory slots follow input width and motif type, not neuron count. 61 sensory inputs support 36 effective dimensions, 14 inputs support about 16 distinguishable cues, and tripling the hidden region from 88 to 280 neurons adds none. Scaling input width together with the hidden region confirms the mechanism by manipulation rather than correlation: at 56 neurons with 14 inputs only 4 to 16 cues survive the 64-step delay, while 90 neurons with 16 inputs hold all 64 cues tested; beyond 90 neurons nothing changes, so the curve is a step and a plateau, not a power law. Growing neurons buys nothing measurable on this substrate; growth would have to add input channels or new motifs to add basins. This is also the one claim in the section that survived repetition across seeds.

The last row of Table 10 is the one place where a change to the node itself pays. Giving each cell a second compartment whose state gates the somatic threshold multiplicatively lets a fixed linear readout solve a context-dependent reversal that no single-compartment node solves, at 1.000 against chance. The control rules out the obvious explanation: when the same rule signal is added to the drive instead of gating the threshold, the state carries a larger separation (8.3 against 0.8) and the readout still fails, because a sum cannot represent a conjunction. Two limits matter. The gain is in rule-time tasks and not in the push–pull failure of Section 5.9, which is a weight problem and stays one; and the gate strength must stay moderate, since $\kappa = 3$ swings the threshold far enough to push every cell to one side and the capability disappears.

6 Discussion

What is shown. The growth contract is implementable on a measured connectome at negligible rebuild cost, its gates fire in the intended order, and the prerequisite measurement is what decides whether structural work is meaningful. Capacity is separately shown to double and quadruple once the generator holds the declared operating point (Section 5.8), which locates the failure of the growth experiments in the substrate's separability rather than in an inability to add cells. Two library-level results are independent of this system: a state-dependent certificate exists where the global bound fails, and a sparse graph is settled at about 1.8 ns per synapse, which makes 10,000 neurons a real-time-scale object rather than a research curiosity.

What is not shown. No growth has been kept, and the prerequisite that blocks it is measured rather than assumed. The negative result is a statement about a pair, not about connectomes: it

says that the declared readout of this wiring under this model at these operating points does not carry the distinction, which is a stronger and more useful statement than a failed attempt to grow.

Growth is not the same problem as capacity sharing. Specialised small circuits handle single tasks, and the library's own connectome lane reports that a reduced graph matched on population means did not match on member behaviour. Retaining transfer while growing requires shared structure, which is where interference appears. The evidence gate and the outcome gate bound the damage of a wrong growth but do not solve sharing.

Value comes from somewhere. A reward channel needs a scale. A scale supplied by the task caps the system at that task; a scale shaped by selection is endogenous but drifts toward whatever is measured. Both are legitimate designs and both must be declared, because the choice determines which measurements are meaningful. The systems that avoid a human score entirely, by making survival and energy the only currency, are the only ones in this space whose objective is not supplied by a person.

The fly lane could not be run here. The strongest test of the mechanism would be a substrate whose prerequisite already passes, and the library's fly lane is the natural candidate: its own measurements report a per-cell-class gain bringing the antennal-lobe code to a cosine of 0.32 where one global gain leaves it at 0.98. That lane is not runnable from the published material: the demonstration repository carries the lane's code, receipts, tests and documentation but no connectome data file, and the library's own connectome document contains no dataset URL. We record this as a checked limitation rather than an omission: the substitute we could run, a worm-graph regime sweep with per-neuron heterogeneity, is reported above and is negative.

Rule families do not mix. A directed measured connectome does not satisfy the equilibrium contrast gradient contract. Structural growth on directed wiring therefore belongs with a predictive-coding or inference-learning rule, or it must accept that its updates are heuristic. Mixing the two under one set of convergence assumptions is not a defensible default, and the prerequisite gate is what keeps a system honest when the rule family it is using does not apply.

7 Reproducibility

Every table above is produced by one file in `probes/`, and `ep4nn/grow.py` runs its own assertions: nothing shrank, every synapse kept an efficacy across every rebuild, and no kept growth went unmeasured. The environment is a virtual environment with NumPy and SciPy installed from a mirror, and the library is a local checkout with `PYTHONPATH` set. The worm connectome is committed with the repository and its hash recorded. No GPU, no external service and no wallet is involved. Settlement numbers depend on the host, so the receipts record operation counts alongside wall-clock times. Three checks are runnable. The artefact set (scan dump, audit dump, four figures) is identical after a full re-run. The growth receipts are byte-identical across two runs, which the receipt format makes possible by carrying counts and no timings. The paper builds byte-identically under a fixed `SOURCE_DATE_EPOCH`. The commands are `repro_check.sh`, `receipt_determinism.sh` and `docs/paper/build.sh`. The figures of this paper are regenerated from the probe artefacts and the scan dump by one command, and the exact input file of every figure is named in its caption.

8 Conclusion

A measured connectome gives structure cheaply and dynamics not at all. The useful question for a system that grows is not how many neurons it has but whether the answer it produces separates the situations that must be separated. The system presented here encodes that question as a

prerequisite, measures it before it grows, and records a refusal to grow as a result in its own right. On the 302-neuron worm graph the prerequisite fails, and reporting that failure is more useful than adding structure to a code that carries nothing. The contract, the receipt format and the state-dependent certificate are the parts we expect to transfer to the next substrate, in particular to connectomes whose motor neurons exist and whose dynamics are heterogeneous.

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Table 10: Nine hypotheses about where the limitation lies, each tested on the measured connectome. "Refuted" means the measured effect is absent or has the wrong sign for the hypothesis. Sources: `probes/probe_manifold.py`, `probe_readout_search.py`, `probe_locomotion.py`, `probe_mapping_layer.py`, `probe_difficulty.py`, `probe_temporal.py`, `probe_temporal_split.py`, `probe_basins.py`, `probe_two_compartment.py`.

hypothesis	measurement	outcome
more neurons buy capability	with 61 sensory inputs the 302-neuron state has participation ratio 36.1 and $k_{90} = 43$	refuted: capacity is not scarce
node scalars buy capability	slope $2 \to 12$ moves the ratio $1.46 \to 1.00$; leak $0, 0.5, 0.9$ are identical; per-neuron jitter 2.17 to 2.49 against a 2.70 baseline	refuted: worse or inert
the declared readout is the defect	a searched ten-cell set raises readout rank $1.88 \to 8.51$, but the worst situation pair subtends 1.3 degrees	refuted for separability
the wiring decides by itself	forward-minus-reverse command never changes sign over twelve conditions, three weightings, five magnitudes; signs are per-cell and correct (27 GABAergic) but a $\gamma = 5$ GABA gain only lifts ρ from 0.39 to 0.83, never reverses the command	refuted: no push–pull
the mapping needs capacity	rank one reaches 100 percent sign accuracy at MAE 0.10; rank 64 lifts the correlation to 0.84 and worsens MAE to 0.12	refuted: rank one suffices
difficulty raises the required rank	0 to 32 nuisance cells move rank-one correlation $0.835 \to 0.806$	refuted: flat
temporal depth needs capacity	after 128 blank steps sign accuracy is 0.74 to 0.79; final states are one-dimensional (PC1 share 1.00) in two clusters (gap ratio 2.71)	refuted, with a mechanism
scale buys memory slots	median slots 16 at 56 neurons, 32 (the measurement ceiling) at 88, 152 and 280, three seeds; with input width scaled alongside, 56 neurons hold 4 to 16 and 90 neurons hold all 64 tested	partially: floor, then saturation at input width
a second compartment buys capability	rule-gated somatic threshold (multiplicative) reaches 1.000 on XOR of stimulus and rule, against 0.29 to 0.47 for the same node with the rule added additively, although the additive arm carries the larger state separation (8.3 against 0.8)	confirmed, for conjunctive tasks only